

COMPUTER NETWORK CONCEPTS

FINAL PROJECT REPORT

Snakes and Ladders Multiplayer Game

Student Information:

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Course: Computer Network Concepts

Project Topic: Snakes and Ladders Multiplayer Game

Programming Language: Java

Development Environment: NetBeans IDE

Server Platform: AWS EC2 Ubuntu Server

Version Control System: Git & GitHub

1. PROJECT DESCRIPTION

This project is a multiplayer Snakes and Ladders game developed using Java socket programming. The application works with a client-server architecture. The server application runs on AWS EC2 cloud infrastructure and manages communication between two different clients.

The project was developed according to the requirements specified in the project document. The game supports:

- Two-player online gameplay
- Start screen and end screen
- Replay system without closing the application
- AWS cloud server connection
- Real-time communication between clients
- Turn-based gameplay system
- Graphical user interface

The project was designed to provide a user-friendly and visually improved experience.

2. PROJECT OBJECTIVE

The main objective of this project is to implement a real-time multiplayer game using computer networking concepts.

During the development process, the following topics were applied:

- TCP socket communication
- Client-server architecture
- Multithreading
- Object serialization
- Real-time data synchronization
- Cloud server deployment
- GUI development using Java Swing

The project also aims to improve practical knowledge related to distributed systems and network programming.

3. TECHNOLOGIES USED

Technology	Purpose
Java	Main programming language
Java Swing	Graphical user interface
TCP Socket Programming	Network communication
AWS EC2	Cloud server hosting
Git & GitHub	Version control
Ubuntu Server	AWS operating system
NetBeans IDE	Software development

4. SYSTEM ARCHITECTURE

The project follows a client-server architecture.

Server Side

The server application:

- Runs on AWS EC2 Ubuntu Server
- Waits for two clients to connect
- Manages game state
- Controls turn order
- Generates dice values
- Synchronizes all player actions
- Broadcasts updates to connected clients

The server application works as a console application.

Client Side

The client application:

- Connects to the AWS server using the public IP address
- Displays the game board
- Allows the player to roll dice
- Receives updates from the server
- Displays current game state
- Shows winner information

5. PROJECT STRUCTURE

The project contains the following packages:

common Package

Message.java

This class is responsible for communication between server and clients.

Functions:

- Sends dice information
- Sends player positions
- Sends player names
- Sends winner information
- Sends game state updates

GameState.java

This class stores the current game state.

Functions:

- Stores player positions
- Stores player names
- Controls snake and ladder positions
- Controls turn system
- Checks game over condition
- Resets game for replay

Server Package

GameServer.java

This class is the main server application.

Functions:

- Starts server socket
- Accepts client connections
- Creates client threads
- Manages multiplayer communication
- Processes player actions
- Synchronizes game state

The server supports two clients simultaneously.

client Package

GameClient.java

This class is the client-side network manager.

Functions:

- Connects to AWS server
- Sends player actions
- Receives server updates
- Updates game interface
- Controls replay system

ui Package

StartScreen.java

This class displays the start screen.

Functions:

- Allows player to enter username
- Starts the game
- Provides user-friendly interface

GameBoard.java

This class displays the main game interface.

Functions:

- Draws the game board
- Displays snakes and ladders
- Displays player pieces
- Displays dice values
- Displays current turn
- Displays winner screen

The interface was designed to be visually improved and user-friendly.

6. GAME RULES

The game follows traditional Snakes and Ladders rules.

Rules:

1. The game is played between two players.
2. Players roll dice in turns.
3. If a player lands on a ladder, the player moves upward.
4. If a player lands on a snake, the player moves downward.
5. The first player reaching square 100 wins the game.
6. Only the active player can roll dice.
7. The game can be restarted after completion.

7. AWS SERVER CONFIGURATION

The server application was deployed on AWS EC2.

AWS Configuration

Setting	Value
Instance Type	t2.micro
Operating System	Ubuntu Server 24.04 LTS
Server Port	5000
Protocol	TCP

Security Group Rules

Port	Purpose
22	SSH Connection
5000	Game Communication

Java Installation Commands

```
sudo apt update
sudo apt install openjdk-17-jdk -y
```

Running the Server

```
javac src/*/*.java
java -cp src server.GameServer
```

8. CLIENT-SERVER COMMUNICATION

The project uses TCP socket communication.

Communication steps:

1. Server starts and waits for players.
2. Clients connect to AWS server IP.
3. Server assigns player IDs.
4. Players send their usernames.
5. Server synchronizes game state.
6. Dice roll requests are sent to server.
7. Server calculates movements.
8. Updated positions are broadcast to all clients.
9. Winner information is sent to both clients.

Object serialization is used for communication.

9. USER INTERFACE DESIGN

The graphical interface was designed using Java Swing.

Interface features:

- Modern game board design
- Snake and ladder graphics
- Player icons
- Dice display
- Turn information
- Winner popup window
- Replay system
- Username support

The interface was designed to improve user experience.

10. GAMEPLAY FLOW

1. Server application starts on AWS.
2. Two clients connect to the server.
3. Players enter their usernames.
4. Start screen closes.
5. Main game board opens.
6. Player 1 starts the game.
7. Dice values are generated by the server.
8. Player positions are synchronized.
9. Winner is determined.
10. Players can replay without restarting application.

11. IMPORTANT NETWORKING CONCEPTS USED

The following networking concepts were implemented in this project:

- TCP communication
- Socket programming
- Cloud server hosting
- Real-time synchronization
- Multithreaded server design
- Client-server architecture
- Data serialization
- Network port management

12. PROBLEMS ENCOUNTERED

During the project development process, several problems were encountered.

AWS Server Connection

Initially, cloud server connection configurations caused communication issues.

Solution:

- Security group rules were configured correctly.
- TCP port 5000 was opened.
- Public IPv4 address was used for communication.

Multiplayer Synchronization

Real-time synchronization between clients required careful message handling.

Solution:

- Game state management was centralized on the server.
- Broadcast communication system was implemented.

Replay System

The replay system initially reset only the client interface.

Solution:

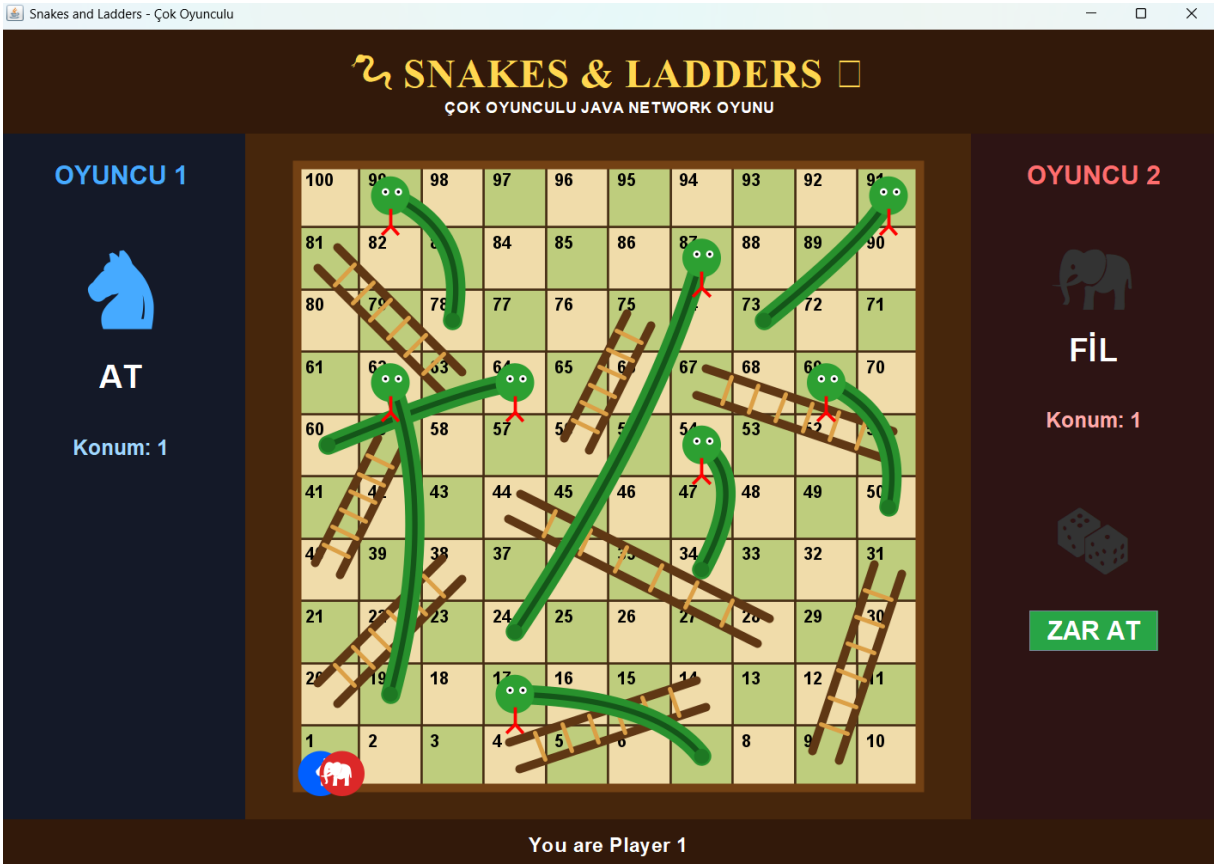
- RESET messages were added to synchronize replay between server and clients.

13. SCREENSHOTS

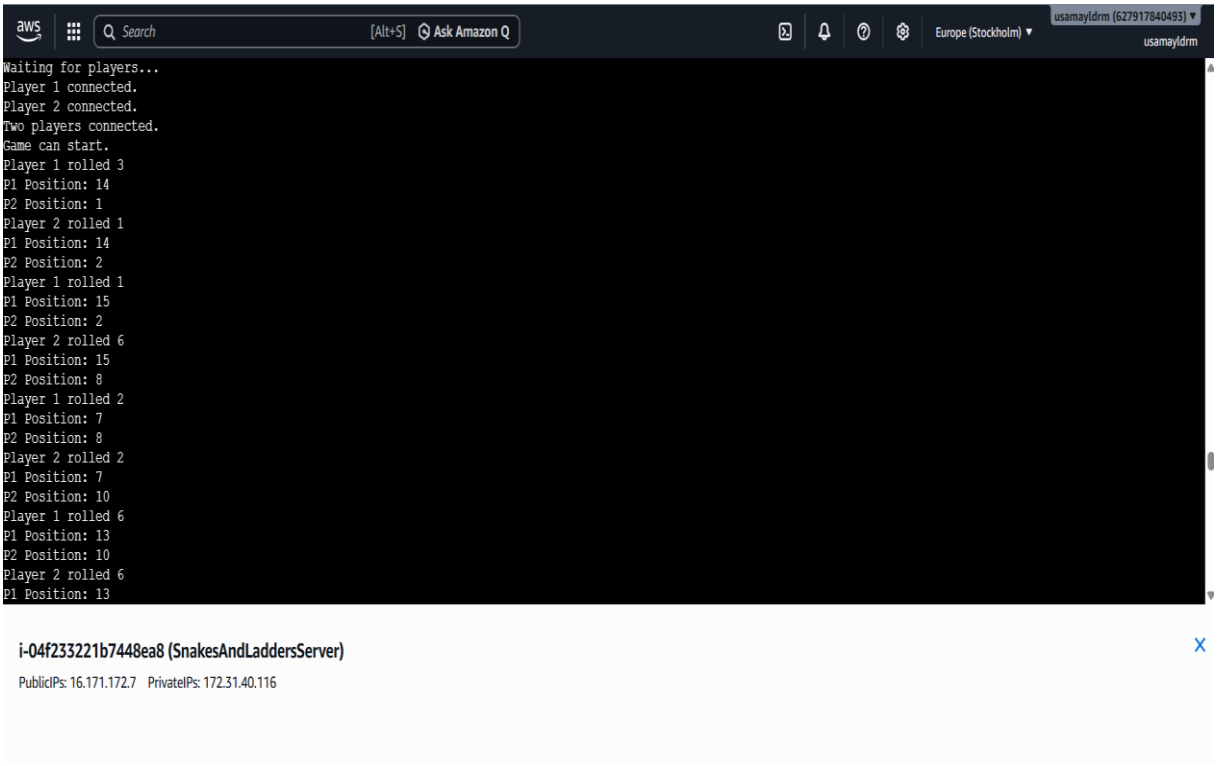
Start Screen



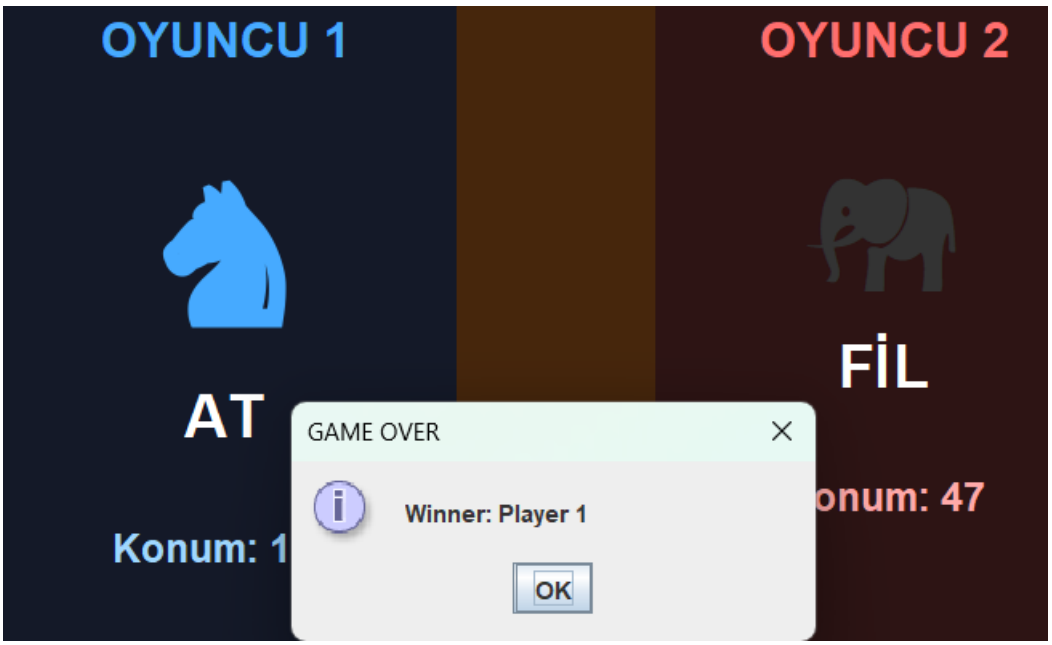
Main Game Board

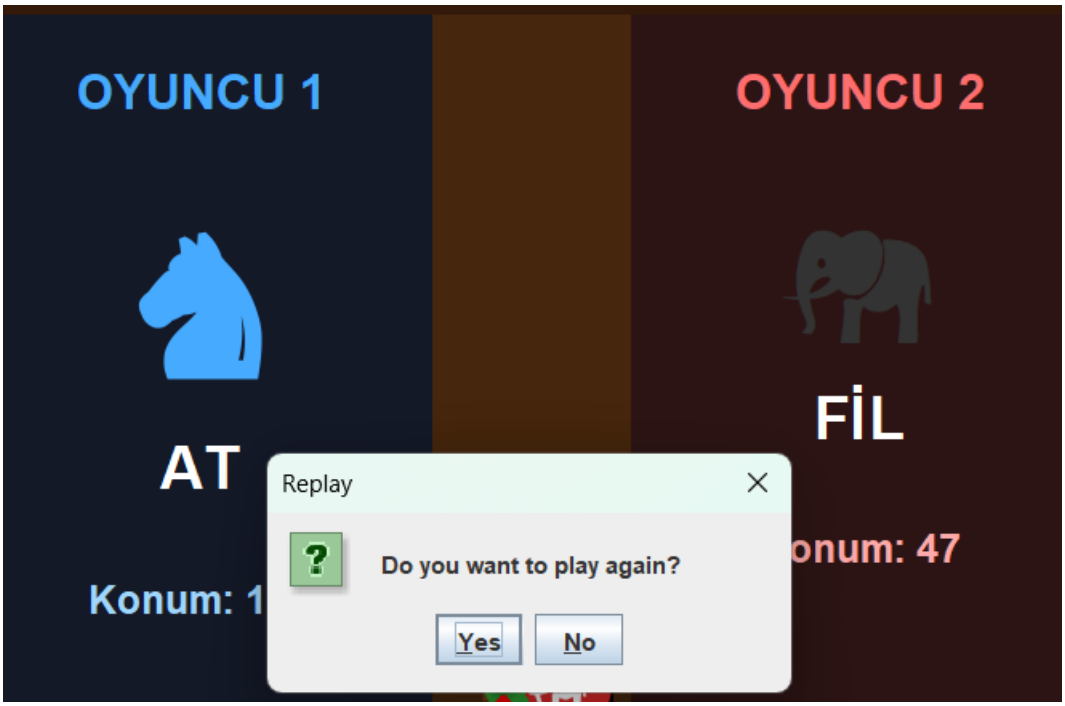


AWS Server Console



Winner Screen





14. CONCLUSION

This project successfully implements a multiplayer Snakes and Ladders game using Java socket programming and AWS cloud infrastructure.

The project satisfies all requirements specified in the project document.

The project provided practical experience in:

- Network programming
- Cloud server management
- Multiplayer game synchronization
- Java GUI development
- Client-server communication

The final system is functional, user-friendly, and deployable.

15. REFERENCES

1. Oracle Java Documentation
2. AWS EC2 Documentation
3. Java Socket Programming Tutorials
4. Java Swing Documentation
5. GitHub Documentation

GitHub Repository

https://github.com/usamayldrm/SnakesAndLaddersNetwork